

# MATHEMATICS-III

## Unit-II: Complex Numbers

Comprehensive Study Notes as per HPU BCA Syllabus

# Topic 1: Introduction to Complex Numbers

## 1.1 What are Complex Numbers?

A **complex number** is a number that can be expressed in the form:

$$z = a + ib$$

where:

- **a** is the **real part** (denoted as  $\text{Re}(z)$ )
- **b** is the **imaginary part** (denoted as  $\text{Im}(z)$ )
- **i** =  $\sqrt{-1}$  (imaginary unit)

**Examples:**

- $z = 3 + 4i \Rightarrow \text{Re}(z) = 3, \text{Im}(z) = 4$
- $z = -2 + 5i \Rightarrow \text{Re}(z) = -2, \text{Im}(z) = 5$
- $z = 7 \Rightarrow \text{Im}(z) = 0$  (purely real)
- $z = 3i \Rightarrow \text{Re}(z) = 0$  (purely imaginary)

## 1.2 Types of Complex Numbers

| Type             | Definition           | Example      |
|------------------|----------------------|--------------|
| Real Number      | $b = 0$              | $z = 5$      |
| Purely Imaginary | $a = 0, b \neq 0$    | $z = 3i$     |
| Complex Number   | $a \neq 0, b \neq 0$ | $z = 2 + 3i$ |
| Zero             | $a = 0, b = 0$       | $z = 0$      |

## 1.3 Equality of Complex Numbers

Two complex numbers  $z_1 = a + ib$  and  $z_2 = c + id$  are equal if:

$$a = c \text{ and } b = d$$

**Example:** If  $3x + 4iy = 6 + 8i$ , then:

$$3x = 6 \Rightarrow x = 2$$

$$4y = 8 \Rightarrow y = 2$$

## Topic 2: Representation of Complex Numbers

### 2.1 Rectangular Representation (Algebraic Form)

$$z = a + ib$$

### 2.2 Polar Form (Trigonometric Form)

$$z = r(\cos \theta + i \sin \theta)$$

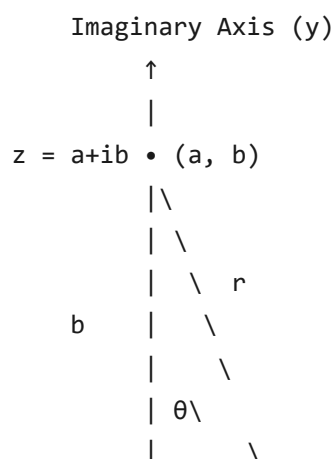
where  $r = |z| = \sqrt{a^2 + b^2}$  and  $\theta = \arg(z) = \tan^{-1}(b/a)$ .

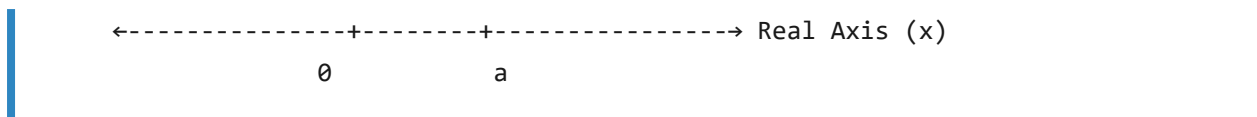
### 2.3 Exponential Form

$$z = r e^{i\theta}$$

### 2.4 Argand Diagram (Geometric Representation)

The **Argand diagram** represents complex numbers as points in a plane. The x-axis is the real axis and the y-axis is the imaginary axis.





| Term                                  | Description                                |
|---------------------------------------|--------------------------------------------|
| <b>Real Axis (x-axis)</b>             | Represents the real part                   |
| <b>Imaginary Axis (y-axis)</b>        | Represents the imaginary part              |
| <b>Point (a, b)</b>                   | Represents the complex number $z = a + ib$ |
| <b>Modulus (r)</b>                    | Distance from origin to the point          |
| <b>Argument (<math>\theta</math>)</b> | Angle made with positive real axis         |

## Topic 3: Algebra of Complex Numbers

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### 3.1 Addition

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If  $z_1 = a + ib$  and  $z_2 = c + id$ , then:

$$z_1 + z_2 = (a + c) + i(b + d)$$

**Example:**  $(3 + 4i) + (2 + 5i) = 5 + 9i$

### 3.2 Subtraction

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$$z_1 - z_2 = (a - c) + i(b - d)$$

**Example:**  $(3 + 4i) - (2 + 5i) = 1 - i$

### 3.3 Multiplication

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$$z_1 \cdot z_2 = (a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

**Example:**  $(3 + 4i)(2 + 5i) = (6 - 20) + i(15 + 8) = -14 + 23i$

### 3.4 Division

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$$z_1 / z_2 = (a + ib)/(c + id) = (ac + bd) / (c^2 + d^2) + i(bc - ad) / (c^2 + d^2)$$

**Example:**

$$\begin{aligned}(3 + 4i) / (2 + 5i) &= (3+4i)(2-5i) / (2^2+5^2) \\ &= (6+20) + i(8-15) / 29 \\ &= (26 - 7i) / 29\end{aligned}$$

## 3.5 Conjugate of a Complex Number

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If  $z = a + ib$ , then the **conjugate** is:

$$z^{-} = a - ib$$

### Properties of Conjugates:

| Property               | Formula                           |
|------------------------|-----------------------------------|
| Conjugate of sum       | $z_1 + z_2 = z_1^{-} + z_2^{-}$   |
| Conjugate of product   | $z_1 z_2 = z_1^{-} \cdot z_2^{-}$ |
| Conjugate of conjugate | $(z^{-})^{-} = z$                 |
| Product with conjugate | $z z^{-} = a^2 + b^2 =  z ^2$     |

## Topic 4: Modulus and Argument

### 4.1 Modulus of a Complex Number

The **modulus** (or absolute value) of  $z = a + ib$  is:

$$|z| = \sqrt{a^2 + b^2}$$

**Properties of Modulus:**

| Property           | Formula                         |
|--------------------|---------------------------------|
| Non-negative       | $ z  \geq 0$                    |
| Zero only for zero | $ z  = 0 \Leftrightarrow z = 0$ |
| Product            | $ z_1 z_2  =  z_1  \cdot  z_2 $ |
| Quotient           | $ z_1 / z_2  =  z_1  /  z_2 $   |
| Power              | $ z^n  =  z ^n$                 |

### 4.2 Argument of a Complex Number

The **argument** of  $z = a + ib$  is the angle  $\theta$  made with the positive real axis:

$$\tan \theta = b / a$$

**In Quadrants:**

| Quadrant | Condition      | Argument                  |
|----------|----------------|---------------------------|
| I        | $a > 0, b > 0$ | $\theta = \tan^{-1}(b/a)$ |



|     |                |                                      |
|-----|----------------|--------------------------------------|
| II  | $a < 0, b > 0$ | $\theta = \pi - \tan^{-1}(b/ a )$    |
| III | $a < 0, b < 0$ | $\theta = -\pi + \tan^{-1}( b / a )$ |
| IV  | $a > 0, b < 0$ | $\theta = -\tan^{-1}( b /a)$         |

**Principal Argument:**  $-\pi < \theta \leq \pi$

## 4.3 Solved Examples

**Example 1:** Find modulus and argument of  $z = 3 + 4i$

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\tan \theta = 4/3 \Rightarrow \theta = \tan^{-1}(4/3) \approx 53.13^\circ$$

$$\text{Answer: } |z| = 5, \arg(z) = 53.13^\circ$$

**Example 2:** Find modulus and argument of  $z = -1 + i$

$$|z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\tan \theta = 1/(-1) = -1$$

Since  $a < 0, b > 0$  (Quadrant II):

$$\theta = \pi - \tan^{-1}(1) = \pi - \pi/4 = 3\pi/4$$

$$\text{Answer: } |z| = \sqrt{2}, \arg(z) = 3\pi/4$$

## Topic 5: Square Root of a Complex Number

### 5.1 Formula

The square root of  $z = a + ib$  is:

$$\sqrt{a + ib} = \pm \left( \sqrt{\frac{|z| + a}{2}} + i \cdot \text{sign}(b) \sqrt{\frac{|z| - a}{2}} \right)$$

where  $|z| = \sqrt{a^2 + b^2}$  and  $\text{sign}(b) = +1$  if  $b > 0$ ,  $-1$  if  $b < 0$ .

### 5.2 Solved Examples

**Example 1:** Find the square root of  $3 + 4i$

$$\begin{aligned} |3 + 4i| &= \sqrt{3^2 + 4^2} = 5 \\ \sqrt{3+4i} &= \pm \left( \sqrt{\frac{(5+3)}{2}} + i \sqrt{\frac{(5-3)}{2}} \right) \\ &= \pm \left( \sqrt{4} + i \sqrt{1} \right) \\ &= \pm (2 + i) \end{aligned}$$

**Example 2:** Find the square root of  $8 - 6i$

$$\begin{aligned} |8 - 6i| &= \sqrt{64 + 36} = 10 \\ \text{Since } b &= -6 < 0: \\ \sqrt{8-6i} &= \pm \left( \sqrt{\frac{(10+8)}{2}} - i \sqrt{\frac{(10-8)}{2}} \right) \\ &= \pm \left( \sqrt{9} - i \sqrt{1} \right) \\ &= \pm (3 - i) \end{aligned}$$

# Topic 6: Cube Roots of Unity

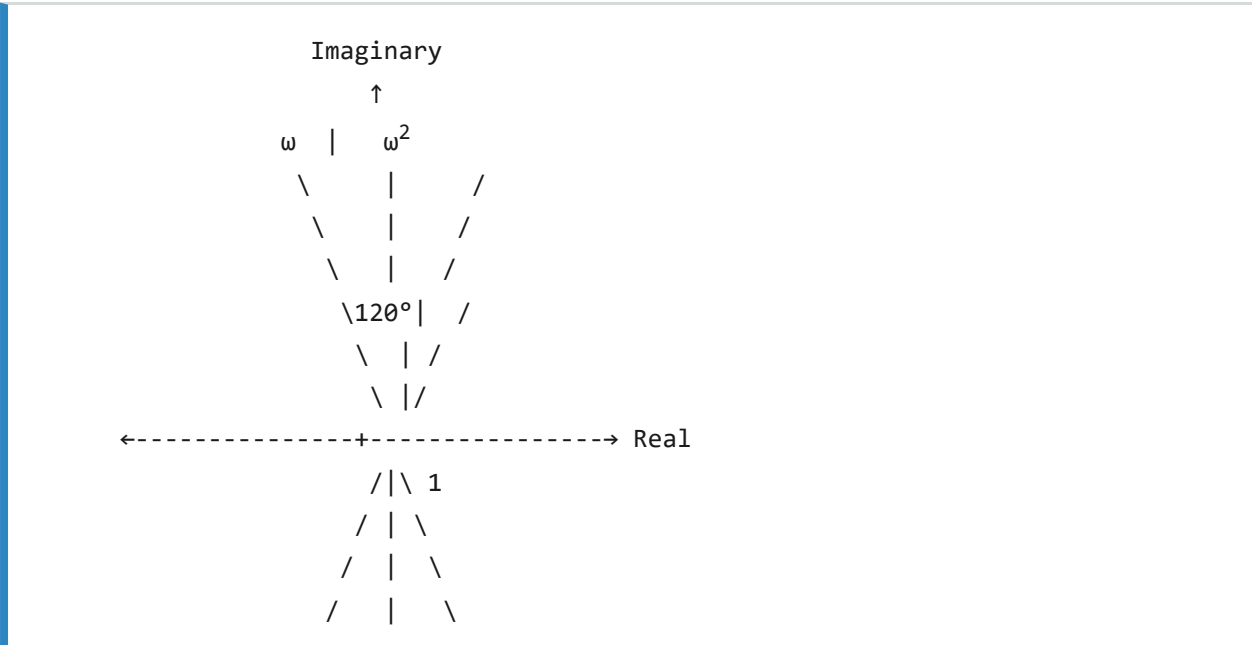
## 6.1 Definition

The **cube roots of unity** are the solutions of  $z^3 = 1$ . The three roots are:

$$1, \omega = (-1 + i\sqrt{3})/2, \omega^2 = (-1 - i\sqrt{3})/2$$

## 6.2 Properties

| Property     | Formula                     |
|--------------|-----------------------------|
| Sum of roots | $1 + \omega + \omega^2 = 0$ |
| Cube         | $\omega^3 = 1$              |
| Power        | $\omega^4 = \omega$         |
| Product      | $\omega \cdot \omega^2 = 1$ |
| Conjugate    | $\omega^- = \omega^2$       |





**Example:** Prove that  $(1 - \omega + \omega^2)(1 + \omega - \omega^2) = 4$

$$1 - \omega + \omega^2 = 1 - \omega + (-1 - \omega) = -2\omega$$

$$1 + \omega - \omega^2 = 1 + \omega - (-1 - \omega) = 2 + 2\omega = 2(1 + \omega) = -2\omega^2$$

$$(1 - \omega + \omega^2)(1 + \omega - \omega^2) = (-2\omega)(-2\omega^2) = 4\omega^3 = 4$$

# Topic 7: Triangle Inequality

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## 7.1 Statement

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For any two complex numbers  $z_1$  and  $z_2$ :

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

## 7.2 Corollaries

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| Inequality          | Formula                                                        |
|---------------------|----------------------------------------------------------------|
| Triangle Inequality | $ z_1 + z_2  \leq  z_1  +  z_2 $                               |
| Reverse Triangle    | $ z_1 - z_2  \geq   z_1  -  z_2  $                             |
| General Form        | $ z_1 + z_2 + \dots + z_n  \leq  z_1  +  z_2  + \dots +  z_n $ |

## Topic 8: De-Moivre's Theorem

### 8.1 Statement

For any real number  $n$  and any complex number  $z = r(\cos \theta + i \sin \theta)$ :

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

$$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$$

### 8.2 Applications

**Example 1:** Find  $(1 + i)^8$

$$\begin{aligned} 1 + i &= \sqrt{2}(\cos \pi/4 + i \sin \pi/4) \\ (1 + i)^8 &= (\sqrt{2})^8(\cos 8\pi/4 + i \sin 8\pi/4) \\ &= 16(\cos 2\pi + i \sin 2\pi) = 16(1 + 0) = 16 \end{aligned}$$

**Example 2:** Find  $(\cos 30^\circ + i \sin 30^\circ)^{12}$

$$\begin{aligned} (\cos 30^\circ + i \sin 30^\circ)^{12} &= \cos(12 \times 30^\circ) + i \sin(12 \times 30^\circ) \\ &= \cos 360^\circ + i \sin 360^\circ = 1 \end{aligned}$$

### 8.3 Finding $n$ th Roots

For  $z = r(\cos \theta + i \sin \theta)$ , the  $n$ th roots are:

$$z^{1/n} = r^{1/n}[\cos((\theta + 2k\pi)/n) + i \sin((\theta + 2k\pi)/n)]$$

where  $k = 0, 1, 2, \dots, n-1$ .

**Example 3:** Find the cube roots of  $8(\cos 60^\circ + i \sin 60^\circ)$

$$r = 8, \theta = 60^\circ$$

$$z^{1/3} = 2[\cos((60^\circ + 360^\circ k)/3) + i \sin((60^\circ + 360^\circ k)/3)]$$

$$k = 0: 2(\cos 20^\circ + i \sin 20^\circ)$$

$$k = 1: 2(\cos 140^\circ + i \sin 140^\circ)$$

$$k = 2: 2(\cos 260^\circ + i \sin 260^\circ)$$

# Topic 9: nth Roots of Complex Numbers

## 9.1 Formula

For  $z = r(\cos \theta + i \sin \theta)$ , the  $n$ th roots are:

$$z^{1/n} = r^{1/n} [\cos((\theta + 2k\pi)/n) + i \sin((\theta + 2k\pi)/n)], k = 0, 1, \dots, n-1$$

## 9.2 Solved Examples

**Example 1:** Find the square roots of  $4 + 4\sqrt{3}i$

$$r = \sqrt{4^2 + (4\sqrt{3})^2} = \sqrt{16 + 48} = \sqrt{64} = 8$$

$$\tan \theta = 4\sqrt{3}/4 = \sqrt{3} \Rightarrow \theta = 60^\circ = \pi/3$$

$$\begin{aligned} k = 0: & \sqrt{8} [\cos(\pi/6) + i \sin(\pi/6)] \\ & = 2\sqrt{2} (\sqrt{3}/2 + i/2) = \sqrt{6} + i\sqrt{2} \end{aligned}$$

$$\begin{aligned} k = 1: & 2\sqrt{2} [\cos(\pi/6 + \pi) + i \sin(\pi/6 + \pi)] \\ & = -\sqrt{6} - i\sqrt{2} \end{aligned}$$

Answer:  $\pm (\sqrt{6} + i\sqrt{2})$

**Example 2:** Find the fourth roots of 16

$$16 = 16(\cos \theta + i \sin \theta), r = 16, \theta = 0^\circ$$

$$z^{1/4} = 2 [\cos(360^\circ k/4) + i \sin(360^\circ k/4)]$$

$$k = 0: 2(\cos 0 + i \sin 0) = 2$$

$$k = 1: 2(\cos 90^\circ + i \sin 90^\circ) = 2i$$

$$k = 2: 2(\cos 180^\circ + i \sin 180^\circ) = -2$$

$$k = 3: 2(\cos 270^\circ + i \sin 270^\circ) = -2i$$

Answer:  $\pm 2, \pm 2i$



## Topic 10: HPU Important Questions

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### Short Answer Questions (2–5 Marks)

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1. Define a complex number and its modulus.
2. What is the Argand diagram?
3. What is the conjugate of a complex number?
4. State De-Moivre's theorem.
5. Find the cube roots of unity.
6. State the triangle inequality for complex numbers.
7. Write the square root formula for a complex number.
8. What is the polar form of a complex number?
9. What is the difference between modulus and argument?
10. Express  $1 + i\sqrt{3}$  in polar form.

### Long Answer Questions (10 Marks)

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1. Find the square root of  $7 + 24i$ .
2. Find the fourth roots of  $8 + 8i\sqrt{3}$ .
3. Solve  $z^3 = 1$  and find cube roots of unity.
4. Prove De-Moivre's theorem and use it to find  $(1 + i)^6$ .
5. Express  $(1+i)^2 / (1-i)$  in standard form.
6. Find the modulus and argument of  $z = (1+i)^3 / (1-i)^2$ .

### Objective Questions (1 Mark)

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1. The imaginary part of  $3 + 4i$  is:  
(a) 3 (b) 4 (c) 5 (d) 7 **Answer: (b)**
2. The modulus of  $3 + 4i$  is:  
(a) 3 (b) 4 (c) 5 (d) 7 **Answer: (c)**
3. The conjugate of  $2 - 3i$  is:  
(a)  $-2 + 3i$  (b)  $2 + 3i$  (c)  $-2 - 3i$  (d)  $2 - 3i$  **Answer: (b)**
4.  $i^2 =$ :  
(a) 1 (b)  $-1$  (c)  $i$  (d)  $-i$  **Answer: (b)**

5. The cube root of unity  $\omega^3 =$ :

- (a) 0 (b) 1 (c)  $-1$  (d)  $i$  **Answer: (b)**

6.  $1 + \omega + \omega^2 =$ :

- (a) 0 (b) 1 (c) 2 (d) 3 **Answer: (a)**

7. De-Moivre's theorem is:

(a)  $(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$

(b)  $(\cos\theta + i \sin\theta)^n = n \cos\theta + i n \sin\theta$

- (c) Both (d) None **Answer: (a)**

8.  $z_1 + z_2 =$ :

- (a)  $z_1^{-1} + z_2^{-1}$  (b)  $z_1^{-1} - z_2^{-1}$  (c)  $z_1^{-1} \cdot z_2^{-1}$  (d) None **Answer: (a)**

# Quick Revision

## Key Formulas Summary

| Formula                    | Description            |
|----------------------------|------------------------|
| $z = a + ib$               | Standard form          |
| $ z  = \sqrt{a^2 + b^2}$   | Modulus                |
| $\arg(z) = \tan^{-1}(b/a)$ | Argument               |
| $z^- = a - ib$             | Conjugate              |
| $z z^- =  z ^2$            | Product with conjugate |

## Polar Form

| Form               | Expression                           |
|--------------------|--------------------------------------|
| Trigonometric      | $z = r(\cos \theta + i \sin \theta)$ |
| Exponential        | $z = r e^{i\theta}$                  |
| $r =  z $          | Modulus                              |
| $\theta = \arg(z)$ | Argument                             |

## De-Moivre's Theorem

| Formula                                                                | Use    |
|------------------------------------------------------------------------|--------|
| $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$      | Powers |
| $z^{1/n} = r^{1/n}[\cos((\theta+2k\pi)/n) + i \sin((\theta+2k\pi)/n)]$ | Roots  |

## Cube Roots of Unity

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| Root       | Value                |
|------------|----------------------|
| 1          | 1                    |
| $\omega$   | $(-1 + i\sqrt{3})/2$ |
| $\omega^2$ | $(-1 - i\sqrt{3})/2$ |

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End of Unit-II Notes — MATHEMATICS-III (HPU BCA)